

24年5318 sample exam questions

Sample exam questions

Question 1. (Multiple choice question)

Select the correct answer.

1. Leave-one-out cross validation is suitable for large data sets.

- a) True
- b) False

Answer:

False. It is not suitable for large databases as it requires n runs where n is the number of examples in the database.

2. The regression line minimizes the sum of the residuals

- a) True
- b) False

Answer:

False. It minimizes the sum of the squared residuals.

3. A single perceptron can solve the XOR problem.

True False

Answer:

False. The XOR problem is linearly inseparable.

Question 2. (Short answer question)

1. Why do we need to apply normalization when using distance-based algorithms such as k-Nearest Neighbor?

Answer: As different attributes are measured on different scales, without normalization the effect of the attributes with smaller scale of measurement will be less significant than those with larger.

2. In linear support vector machines, we use dot products both during training and during classification of a new example. What vectors are these products of?

During training:

During classification of new example:

During classification of a new example: The new example and the support vectors

3. List one disadvantage of applying a multi-layer perceptron neural network to perform handwritten digits image classification.

Answer:

The network may be sensitive to change of position, size and angle in the images.
(Only 1 of these is sufficient – the question asks about 1 disadvantage)

Calculation (problem solving) questions

Question 3. Decision tree

Given is the following training data where *location*, *weather* and *expensive* are the features and *holiday* is the class.

location	weather	expensive	holiday
nice	sunny	Y	good
nice	sunny	N	bad
boring	rainy	Y	good
boring	sunny	N	bad
nice	rainy	Y	good
boring	rainy	N	good
boring	rainy	N	good

- What is the entropy of this set of training examples with respect to the class?
- We would like to build a decision tree using information gain. Which attribute will be selected as a root of the tree? Show your calculations.

You may use this table:

x	y	$-(x/y) * \log_2(x/y)$	x	y	$-(x/y) * \log_2(x/y)$
1	2	0.50	1	6	0.43
1	3	0.53	5	6	0.22
2	3	0.39	1	7	0.40
1	4	0.5	2	7	0.52
3	4	0.31	3	7	0.52
1	5	0.46	4	7	0.46
2	5	0.53	5	7	0.35
3	5	0.44	6	7	0.19
4	5	0.26			

Solution:

We need to calculate the information gain of each of the three attribute and select the one with the highest information gain.

a) $H(S) = I(5/7, 2/7) = -(5/7)\log(5/7) - (2/7)\log(2/7) = 0.35 + 0.52 = 0.87$ bits

b) Split on *location*:

$H(S_{\text{nice}}) = I(2/3, 1/3) = -(2/3)\log(2/3) - (1/3)\log(1/3) = 0.39 + 0.53 = 0.92$ bits

$H(S_boring) = I(3/4, 1/4) = -(3/4)\log(3/4) - (1/4)\log(1/4) = 0.31 + 0.5 = 0.81$ bits
 $H(S, location) = 3/7 * 0.92 + 4/7 * 0.81 = 0.86$ bits
 $gain(location) = 0.87 - 0.86 = 0.01$ bits

split on **weather**:

$entropy(S_sunny) = I(1/3, 2/3) = -(1/3)\log(1/3) - (2/3)\log(2/3) = 0.53 + 0.39 = 0.92$ bits
 $entropy(S_rainy) = I(4/4, 0/4) = -(4/4)\log 1 - (0/4)\log 0 = 0$ bits
 $entropy(S, weather) = 3/7 * 0.92 + 4/7 * 0 = 0.39$ bits
 $gain(weather) = 0.87 - 0.39 = 0.48$ bits

split on **expensive**:

$entropy(S_Y) = I(3/3, 1/3) = -(3/3)\log 1 - (0/3)\log 0 = 0$ bits
 $entropy(S_N) = I(2/4, 2/4) = -(2/4)\log(2/4) - (2/4)\log(2/4) = 0.5 + 0.5 = 1$ bit
 $entropy(S, expensive) = 3/7 * 0 + 4/7 * 1 = 0.57$ bits
 $gain(expensive) = 0.87 - 0.57 = 0.30$ bits

As **weather** has the highest information gain, it will be selected as the root of the tree (i.e. the first attribute to split on).

Question 4. Naïve Bayes

Given is the following training data where **location**, **weather**, **companion** and **expensive** are the features and **holiday** is the class.

location	weather	companion	expensive	holiday
nice	sunny	annoying	Y	good
nice	sunny	annoying	N	bad
boring	rainy	great	Y	good
boring	sunny	great	Y	bad
nice	rainy	great	Y	good
boring	rainy	annoying	N	good
boring	rainy	great	N	good

Use Naïve Bayes to predict the value of **holiday** for the following new example, showing your calculations:
location=boring, weather=sunny, companion=annoying, expensive=Y.

Solution:

$E = location=boring, weather=sunny, companion=annoying, expensive=Y$

$E1$ is $location=boring$, $E2$ is $weather=sunny$, $E3$ is $companion=annoying$, $E4$ is $expensive=Y$

We need to compute $P(good|E)$ and $P(bad|E)$ and compare them.

$$P(good|E) = \frac{P(E_1|good)P(E_2|good)P(E_3|good)P(E_4|good)P(good)}{P(E)}$$

$$P(bad|E) = \frac{P(E_1|bad)P(E_2|bad)P(E_3|bad)P(E_4|bad)P(bad)}{P(E)}$$

$$P(good) = 5/7$$

$$P(bad) = 2/7$$

$$P(E1|good) = P(location=boring|good) = 3/5$$

$$P(E1|bad) = P(location=boring|bad) = 1/2$$

$$P(E2|good) = P(weather=sunny|good) = 1/5$$

$$P(E2|bad) = P(weather=sunny|bad) = 2/2$$

$$P(E3|good) = P(companion=annoying|good) = 2/5$$

$$P(E3|bad) = P(companion=annoying|bad) = 1/2$$

$$P(E4|good)=P(\text{expensive}=Y|good)=3/5$$

$$P(E4|bad)=P(\text{expensive}=Y|bad)=1/2$$

$$P(good|E) = \frac{\frac{3}{5} \cdot \frac{1}{5} \cdot \frac{2}{5} \cdot \frac{3}{5} \cdot \frac{5}{7}}{P(E)} = \frac{\frac{18}{875}}{P(E)} = \frac{0.02}{P(E)}$$

$$P(bad|E) = \frac{\frac{1}{2} \cdot \frac{2}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{2}{7}}{P(E)} = \frac{\frac{1}{28}}{P(E)} = \frac{0.04}{P(E)}$$

$P(bad|E) > P(good|E) \Rightarrow$ Naïve Bayes predicts **holiday=bad** for the new example.

Question 5. 1R

Given the training data in the table below where **credit history**, **debt**, **deposit** and **income** are attributes and **risk** is the class, predict the class of the following new example using the 1R algorithm: *credit history=unknown, debt=low, deposit=none, income=average*. If needed, settle ties by random selection. Show your calculations.

credit history	debt	deposit	income	risk
bad	high	none	low	high
unknown	high	none	average	high
unknown	low	none	average	moderate
unknown	low	none	low	high
unknown	low	none	high	low
unknown	low	adequate	high	low
bad	low	none	low	high
bad	low	adequate	high	moderate
good	low	none	high	low
good	high	adequate	high	low
good	high	none	low	high
good	high	none	average	moderate
good	high	none	high	low
bad	high	none	average	high

Solution:

1. Attribute **credit history**

bad: 0 low, 1 moderate, 3 high $\Rightarrow$ risk=high, errors: 1/4

unknown: 2 low, 1 moderate, 2 high $\Rightarrow$ risk=low, errors: 3/5

good: 3 low, 1 moderate, 1 high $\Rightarrow$ risk=low, errors: 2/5

total errors: 6/14

2. Attribute **debt**

high: 2 low, 1 moderate, 4 high $\Rightarrow$ risk=high, errors: 3/7

low: 3 low, 2 moderate, 2 high $\Rightarrow$ risk=low, errors: 4/7

total errors: 7/14

3. Attribute **deposit**

none: 3 low, 2 moderate, 6 high $\Rightarrow$ risk=high, errors: 5/11

adequate: 2 low, 1 moderate, 0 high $\Rightarrow$ risk=low, errors: 1/3

total errors: 6/14

4. Attribute **income**

low: 0 low, 0 moderate, 4 high => risk=high, errors: 0/4

average: 0 low, 2 moderate, 2 high => risk=moderate, errors: 2/4 (tie - random selection between moderate and high)

high: 5 low, 1 moderate, 0 high => risk=low, errors: 1/6

total errors: 3/14

The rule based on *income* has the minimum number of errors => 1R produces the following rule:

if income=low then risk=high

else if income=average then risk=moderate

else if income=high then risk=low

The new example has *income=average* and hence will be classified as *risk=moderate*.

Question 6. Perceptron

Given is the following training set:

	input	output
ex. 1:	1 0 0	1
ex. 2:	0 1 1	0
ex. 3:	1 1 0	1
ex. 4:	1 1 1	0
ex. 5:	0 0 1	0

a) Train a perceptron **with a bias** on this training set. Assume that all initial weights (including the bias of the neuron) are 0. Show the set of weights (including the bias) at the end of the first epoch. Apply the examples in the given order.

Recall that the perceptron uses a step function defined as:

$\text{step}(n) = 1$, if $n \geq 0$
 $= 0$, otherwise.

Solution:

a) starting point: $w=[0\ 0\ 0]$, $b=0$

applying ex.1: $p_1=[1\ 0\ 0]$, $t_1=1$

$a=\text{step}([0\ 0\ 0][1\ 0\ 0]+0)=\text{step}(0)=1$

$e=t_1-a=1-1=0$

$w^{\text{new}}=[0\ 0\ 0]+0[1\ 0\ 0]=[0\ 0\ 0]$

$b^{\text{new}}=0+0=0$

i.e. no change in w and b as $e=0$

applying ex.2: $p_2=[0\ 1\ 1]$, $t_2=0$

$a=\text{step}([0\ 0\ 0][0\ 1\ 1]+0)=\text{step}(0)=1$

$e=t_2-a=0-1=-1$

$w^{\text{new}}=[0\ 0\ 0]+(-1)[0\ 1\ 1]=[0\ -1\ -1]$

$b^{\text{new}}=0+(-1)=-1$

i.e. w and b have been updated

Similarly:

After applying ex. 3: $w^{\text{new}}=[1\ 0\ -1]$, $b^{\text{new}}=0$

After applying ex. 4: $w^{\text{new}}=[0\ -1\ -2]$, $b^{\text{new}}=-1$

After applying ex. 5: no change
end of epoch 1

Weight vector and bias at the end of epoch 1: $w = [0 \ -1 \ -2]$, $b = -1$

Question 7. K-means clustering

Suppose that we are given 7 examples to cluster: A, B, C, D, E, F and G. The distance between them is given by the following matrix:

	A	B	C	D	E	F	G
A	0	10	2	1	12	5	4
B	10	0	4	3	6	23	7
C	2	4	0	5	9	14	19
D	1	3	5	0	1	7	4
E	12	6	9	1	0	2	18
F	5	23	14	7	2	0	6
G	4	7	19	4	18	6	0

Run the **k-means** algorithm to group these examples into 2 clusters for 1 epoch. The initial centroids are A and B. Show the resulting clusters.

Solution:

We need to find the distance between C, D, E, F and G to the two initial centroids (A and B), and assign each of them to the cluster of the closest centroid.

Note that the matrix is symmetric as $d(X,Y)=d(Y,X)$.

Let K1 is the cluster of A and K2 is the cluster of B.

C: $d(C, A)=2$, $d(C, B)=4 \Rightarrow$ C is assigned to K1 (this comparison is shown in the matrix below)

D: $d(D, A)=1$, $d(D, B)=3 \Rightarrow$ D is assigned to K1

E: $d(E, A)=12$, $d(E, B)=6 \Rightarrow$ E is assigned to K2

F: $d(F, A)=5$, $d(F, B)=23 \Rightarrow$ F is assigned to K1

G: $d(G, A)=4$, $d(G, B)=7 \Rightarrow$ G is assigned to K1

Clusters at the end of epoch 1:

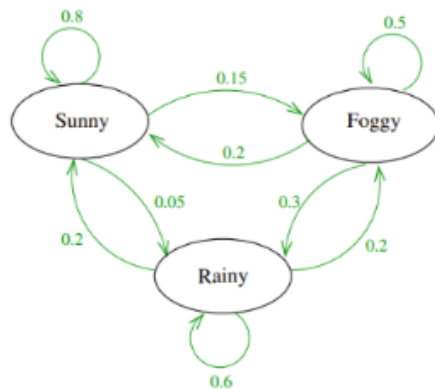
$K1=\{A,C,D,F,G\}$, $K2=\{B,E\}$

(This matrix shows the distance comparison and example assignment:)

	A	B	C	D	E	F	G
A	0	10	2	1	12	5	4
B	10	0	4	3	6	23	7
C	2	4	0	5	9	14	19
D	1	3	5	0	1	7	4
E	12	6	9	1	0	2	18
F	5	23	14	7	2	0	6
G	4	7	19	4	18	6	0

Question 8. Markov models

Given is the following Markov model for the weather in Sydney:



a) Given that today the weather is *Sunny*, what is the probability that it will be *Sunny* tomorrow and *Rainy* the day after tomorrow, i.e. what is the probability $P(\pi_3 = \text{Rainy}, \pi_2 = \text{Sunny} | \pi_1 = \text{Sunny})$?

Hint: $P(A,B|C) = P(A|B,C) P(B|C)$

b) If the weather yesterday was *Rainy*, and today is *Foggy*, what is the probability that tomorrow it will be *Sunny*?

For both questions, briefly show your calculations.

Solution:

a)

$$\begin{aligned} P(\pi_3 = \text{Rainy}, \pi_2 = \text{Sunny} | \pi_1 = \text{Sunny}) &= \\ &= P(\pi_3 = \text{Rainy} | \pi_2 = \text{Sunny}, \pi_1 = \text{Sunny}) * P(\pi_2 = \text{Sunny} | \pi_1 = \text{Sunny}) = \text{(rule from hint)} \\ &= P(\pi_3 = \text{Rainy} | \pi_2 = \text{Sunny}) * P(\pi_2 = \text{Sunny} | \pi_1 = \text{Sunny}) = \text{(Markov assumption)} \\ &= 0.05 * 0.8 = 0.04 \end{aligned}$$

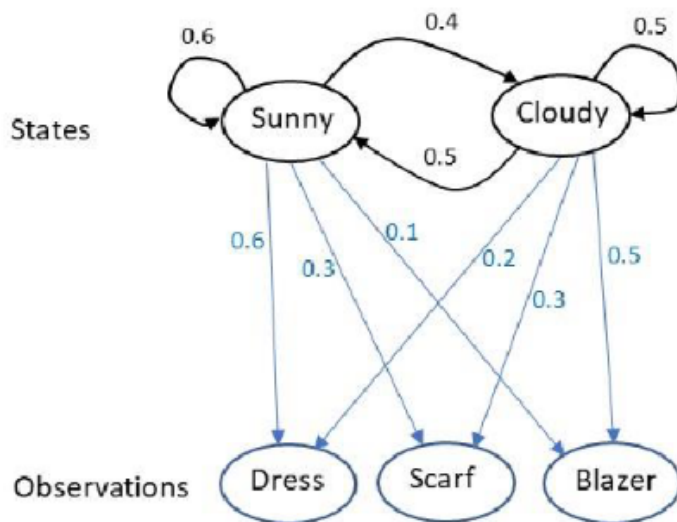
b)

$$\begin{aligned} P(\pi_3 = \text{Sunny} | \pi_2 = \text{Foggy}, \pi_1 = \text{Rainy}) &= \\ &= P(\pi_3 = \text{Sunny} | \pi_2 = \text{Foggy}) = \text{(Markov assumption)} \\ &= 0.2 \end{aligned}$$

Question 9. Hidden Markov models

Julia tested positive to COVID and had to quarantine at home for several days. Her friend Nicole came to bring her food every day. We don't know what the weather was on the quarantine days but we know the type of clothing Nicole wore and it provides evidence about the weather.

The following Hidden Markov Model models the situation. The initial state probabilities are: $A_0(\text{Sunny})=0.5$ and $A_0(\text{Cloudy})=0.5$.



Suppose that on the first quarantine day Nicole wore a dress and on the second she wore a blazer.

- What is the probability of the observation sequence?
- What is the most likely sequence of hidden states?

Briefly show your calculations.

Solution:

States: Sunny (S) and Cloudy (C)

Observations: Dress (D), Scarf (Sc) and Blazer (B)

Observation sequence: $X=\text{Dress (D), Blazer (B)}$

- This is HMM problem 1. We can use the Forward algorithm to solve it.**

Step 1: Initialization

Day 1: D observed

$$V_S(1) = A_0(S)e_S(D) = 0.5 \cdot 0.6 = 0.3$$

$$V_C(1) = A_0(C)e_C(D) = 0.5 \cdot 0.2 = 0.1$$

Step 2: Iteration

Day 2: B observed

$$\begin{aligned} V_S(2) &= e_S(B) \cdot (V_S(1)a_{SS} + V_C(1)a_{CS}) = \\ &= 0.1 \cdot (0.3 \cdot 0.6 + 0.1 \cdot 0.5) = 0.1 \cdot (0.18 + 0.05) = 0.023 \end{aligned}$$

$$V_C(2) = e_C(B) * (V_S(1)a_{SC} + V_C(1)a_{CC}) = \\ = 0.5 * (0.3 * 0.4 + 0.1 * 0.5) = 0.5 * (0.12 + 0.05) = 0.085$$

Step 3: Termination

$$P(X) = P(D, B) = 0.023 + 0.085 = 0.108$$

The probability of the observation sequence Dress, Blazer is 0.108.

b) This is HMM problem 2. We can use the Viterbi algorithm to solve it.

Step 1: Initialization

Day 1: D observed

$$V_S(1) = A_0(S)e_S(D) = 0.5 * 0.6 = 0.3 \\ V_C(1) = A_0(C)e_C(D) = 0.5 * 0.2 = 0.1$$

Step 2: Iteration

Day 2: B observed

$$V_S(2) = e_S(B) * \max(V_S(1)a_{SS}, V_C(1)a_{CS}) = \\ = 0.1 * \max(0.3 * 0.6, 0.1 * 0.5) = 0.1 * 0.3 * 0.6 = 0.018 \\ Ptr_S(2) = \operatorname{argmax}(0.3 * 0.6, 0.1 * 0.5) = 1, \text{ i.e. } S$$

$$V_C(2) = e_C(B) * \max(V_S(1)a_{SC}, V_C(1)a_{CC}) = \\ = 0.5 * \max(0.3 * 0.4, 0.1 * 0.5) = 0.5 * 0.3 * 0.4 = 0.06 \\ Ptr_C(2) = \operatorname{argmax}(0.3 * 0.4, 0.1 * 0.5) = 1, \text{ i.e. } S$$

Step 3: Termination and trace-back

$$\text{Final state} = \operatorname{argmax}(V_S(2), V_C(2)) = \operatorname{argmax}(0.018, 0.06) = 2, \text{ i.e. } C$$

Trace-back through pointers:

$$Ptr_C(2) = S$$

Hence, the most likely sequence of hidden states is S, C (Sunny, Cloudy)

